



Final Report

Title: Neuropharmacological Profile (NPP) of LT3114
in Rats (GLP)

**Calvert
Study No.:** 0200RL35.001

**Testing
Facility:** Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

Date: 23 Dec 2014

LABORATORY

Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447 USA
Phone: 570.586.2411
Fax: 570.586.3450
www.calvertlabs.com

CORPORATE

1225 Crescent Green Dr.
Suite 115
Cary, NC 27518
Phone: 919.854.4453
Fax: 919.854.2860
www.calvertholdings.com

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II. List of Abbreviations

ANOVA	Analysis of Variance
CFR	Code of Federal Regulations
CO ₂	Carbon dioxide
CS	Calibration Standard
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
IACUC	Institutional Animal Care and Use Committee
ICH	International Conference on Harmonisation
ILAR	Institute for Laboratory Animal Research
LLOQ	Lower Limit of Quantification
NPP	Neuropharmacological Profile
OECD	Organisation for Economic Co-operation and Development
pd	post-dose
QAU	Quality Assurance Unit
QC	Quality Control
q.s.	Quantity Sufficient
SEM	Standard Error of the Mean
SOP	Standard Operating Procedure
USDA	United States Department of Agriculture
%CV	Coefficient of Variation

III. Regulatory Compliance and Signatures

Calvert Study No.: 0200RL35.001

Title: Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

A. Regulatory Compliance

1. Calvert Laboratories-Conducted Study Elements

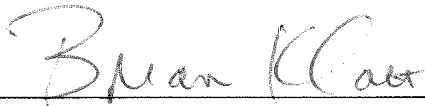
All parts of this study were performed according to SOPs and the protocol and conformed to the U.S. FDA GLP regulations in effect at the time of study conduct (Good Laboratory Practice Regulations, Title 21 Code of Federal Regulations Part 58), in accordance with OECD Principles of Good Laboratory Practice as revised on 26th November, 1997 [C(97)186/Final].

2. Guidelines for Study Design

FDA Guidance for Industry "S7A Safety Pharmacology Studies for Human Pharmaceuticals (ICH)" issued July 2001

B. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained. Protocol deviation(s) can be found in Appendix III.



Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.



Date

C. Signatures of Other Responsible Personnel

Scientific Oversight and Report Review



Scientific Management
Calvert Laboratories, Inc.



Date

IV. Quality Assurance Unit Statement

Calvert Study No.: 0200RL35.001

Title: Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

The following inspections were made in accordance with appropriate Calvert Standard Operating Procedures.

A. Protocol Inspection

Date of Inspection	Report to Study Director	Report to Management
11 Aug 2014	11 Aug 2014	11 Aug 2014

B. Protocol Amendment

Amendment	Date of Inspection	Report to Study Director	Report to Management
1	21 Aug 2014	22 Aug 2014	22 Aug 2014
2	3 Nov 2014	3 Nov 2014	3 Nov 2014

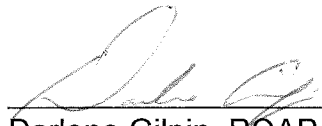
C. Procedures

Inspected Phase	Date of Inspection	Report to Study Director	Report to Management
Dosing	11 Aug 2014	11 Aug 2014	11 Aug 2014

D. Data/Draft Report Inspection

Inspected Phase	Date of Inspection	Report to Study Director	Report to Management
Raw Data/Draft Report #1	11, 12 & 15 Sep 2014	15 Sep 2014	15 Sep 2014
Draft Report #2	3 Nov 2014	3 Nov 2014	3 Nov 2014
Final Report	22 Dec 2014	22 Dec 2014	22 Dec 2014

This report has been reviewed by the Quality Assurance Unit, employing methods detailed in appropriate Calvert Standard Operating Procedures. The results constitute an accurate representation of recorded data. Any data supplied by the Sponsor/Subcontractor were not audited by the Quality Assurance Unit at Calvert. [Refer to individual Principal Investigator reports included in the appendices of this report for Quality Assurance Statements from test sites].



Darlene Gilpin, RQAP-GLP
Director, Quality Assurance
Calvert Laboratories, Inc.

23 Dec 2014
Date

V. Summary

A. Title

Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

B. Objective

The purpose of this study was to evaluate the potential neuropharmacological effects of LT3114 in male and female rats.

C. Methods

Three groups of ten Sprague Dawley rats (five/sex) were administered LT3114 at doses of 10, 30 and 50 mg/kg by slow intravenous injection (~1 minute). An additional group of ten rats (five/sex) was administered the vehicle (LT3114 Diluent) at 5 mL/kg.

The rats were observed for apparent neuropharmacological or clinical signs at 15, 30 and 45 minutes and 1, 2, 3, 4, and 24 hours following treatment. Body temperatures were measured at 30 minutes (± 5 minutes) following dose administration.

D. Results

No apparent neuropharmacological signs were observed through 24 hours post-dose in any rats receiving the vehicle at 5 mL/kg or LT3114 at doses of 10, 30 or 50 mg/kg by intravenous injection.

Body temperature was not affected by any of the dose levels at 30 minutes following dosing.

E. Conclusion

The intravenous administration of LT3114 at 10, 30 and 50 mg/kg did not produce any apparent neuropharmacological effects in male or female rats. Body temperature was not affected by the intravenous administration of LT3114 at doses up to 50 mg/kg.

VI. General Information

A. Key Study Dates

Study Initiation Date:	1 Aug 2014
Animal Receipt Date:	6 Aug 2014
Experimental Start Date:	11 Aug 2014
Experimental Completion Date:	12 Aug 2014

B. Responsible Personnel Testing Facility

Study Director:	Brian Klatt, B.S.
Primary Technicians:	Brian Klatt, B.S. Jennifer Shenise, A.S., ALAT Maura Boyarsky
Pharmacy Technicians:	Katelynn Henrich, B.S.

C. Contributing Scientists

Principal Investigator	
Dosing Formulation Analyses:	Victor J. Johnson, Ph.D. Grants & Contracts PI/Study Director Burleson Research Technologies 120 First Flight Lane Morrisville, NC 27560 Phone: (919) 719-2500 Fax: (866) 971-2128 Email: vjohnson@brt-labs.com
Study Monitor:	Marlene W. Modi, Ph.D. MWM Consulting Group, Inc. for Lpath Inc. 14 Joseph Court Monmouth Junction, NJ 08852 Phone: (732) 274-9562 Mobile: (732) 822-6101 E-mail: mwmconsultinggroup@gmail.com
Sponsor Representative:	Gary Woodnutt, PhD

Senior Vice President, Research
Lpath Inc
4025 Sorrento Valley Blvd
San Diego, CA 92121
Phone: (858) 678-0800 ext 112
Email: gwoodnutt@lpath.com

D. Objective

The purpose of this study was to evaluate the potential neuropharmacological effects of LT3114 in male and female rats.

E. Experimental Design Overview

1. General Description

Three groups of ten Sprague Dawley rats (five/sex) were administered LT3114 at doses of 10, 30 and 50 mg/kg by slow intravenous injection (~1 minute). An additional group of ten rats (five/sex) was administered the vehicle (LT3114 Diluent) at 5 mL/kg.

The rats were observed for apparent neuropharmacological or clinical signs at 15, 30 and 45 minutes and 1, 2, 3, 4, and 24 hours following treatment. Body temperatures were measured at 30 minutes (± 5 minutes) following dose administration.

VII. Materials and Methods

A. Test Article, Vehicle and Preparation

1. Test Article

Identification:	LT3114 (Bulk Drug Substance, Antibody LT-3114)
Lot No.:	103948
Physical Description:	Clear, colorless liquid
Storage Conditions:	Refrigerated (4-7°C) and protected from light.

Characterization and
Stability:

A Certificate of Analysis was provided and is included in Appendix I. Three month stability is provided in Appendix I.

2. Vehicle

Identification:	LT3114 Diluent [24.3 mM Citrate, 51.4 mM Sodium Phosphate, 300 mM D-trehalose pH 5.0 (DF Buffer)]
Lot/Batch No.:	104011-001
Physical Description:	Clear colorless liquid
Storage Conditions:	4-7°C
Preparation:	The diluent and Polysorbate 80 were removed from the refrigerator and retained at room temperature for at least 30 minutes prior to preparation. All procedures were carried out in a cleaned laminar flow hood using aseptic techniques whenever possible. For each bag of diluent, 2 mL of Polysorbate 80 was measured using a sterile 3 mL syringe and sterile 18 gauge needle and injected directly into the bag of vehicle. The bag was gently inverted until the solution appeared homogeneous. Once the bags were prepared, they were removed from the cleaned laminar flow hood and returned to their appropriate storage condition.

3. Test Article Preparation

The Sponsor was provided the test article already formulated at 32.1 mg/ml in sterile containers. Diluent was also provided in separate sterile containers. Diluent and test article were allowed to come to room temperature before the formulation was prepared. The formulation was not shaken. Group 1 control animals received the diluent. When necessary depending on the dose level, doses were prepared by dilution using the

provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dose preparation, 2 mL of the 10% Polysorbate 80 solution provided, was added aseptically to each liter of formulation buffer, also supplied, and inverted to ensure distribution throughout the volume of buffer. This diluent was then used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

The appropriate volume of LT3114 stock solution was diluted in the appropriate volume of vehicle to achieve final nominal concentration dosing concentrations on a mg/ml basis as follows:

Dose concentration	Amount of LT3114 32.1 mg/mL Stock	Volume of Vehicle	Description (pH)
2 mg/mL	1.31 mL	19.69 mL	Clear, colorless liquid (5)
6 mg/mL	3.93 mL	17.07 mL	Clear, colorless liquid (5)
10 mg/mL	6.54 mL	14.46 mL	Clear, colorless liquid (5)

The appropriate amount of LT3114 stock solution was measured using appropriately sized sterile syringes and dispensed into appropriately sized sterile, clear glass vials. Vehicle was measured using appropriately sized sterile syringes and added to the vial. The vial was gently swirled until the material appeared homogeneous. Formulation samples were taken. The vial was sealed with a sterile rubber stopper and crimp cap and kept at room temperature until used for dosing. pH was taken using Litmus paper.

Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

4. Dose Formulation Samples

On the day of dosing, duplicate 1.0 mL samples were collected from the top, middle and bottom of each formulation to confirm concentration and homogeneity. Duplicate samples (1.0 mL) from the middle of the vehicle were collected to confirm the absence of test article. These samples were stored at 2-5°C until shipment for dose formulation analysis.

5. *Dose Formulation Analysis*

Dose formulation samples were shipped to (on wet ice) and analyses was performed under the direction of:

Victor J. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

Good Laboratory Practice (GLP) regulations require that all formulated test articles be appropriately verified for concentration, homogeneity and stability.

Targeted Total Number of

Formulation Samples: 20 Collected, 10 initial shipment (one set shipped for analysis and the second set retained at Calvert as a backup)

Sample Storage:

Conditions: Refrigerated (2-8°C)

Sample Ship Date:

Upon approval of analytical laboratory (3 Sep 2014)

Testing Requirements:

Concentration and homogeneity

Analytical Methods:

BRT 20140629-A

Automated Data

Collection Systems:

Spectramax 340 microplate spectrophotometer and/or Spectramax M2e microplate spectrophotometer

Analysis of the ELISA absorbance data was performed with Softmax Pro v2.2.1.

Analysis of dose formulation acceptance criteria was performed using Microsoft Excel v2013

Dose formulation stability has been conducted under BRT 20140629-A.

If any samples are re-analyzed, the Study Director was notified and details were included in the analytical report.

All Quality Assurance Unit activities and Quality Assurance Unit reports for the analytical analyses were sent to the Calvert Study Director and Calvert Management. All protocol, SOP and GLP deviations associated with the analytical analyses were reported to the Calvert Study Director. The final analytical report includes a GLP compliance statement and a Quality Assurance Unit statement which has the dates of any inspections and the dates the QAU reports were sent to the Principal Investigator, Study Director, and Study Director's Management.

A report of the sample analysis procedures and results was provided to the Study Director and is included in Appendix II. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.0 – 2.1 mg/mL, 5.7 – 6.5 mg/mL, and 10.6 – 10.9 mg/mL for the 2.0, 6.0, and 10.0 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 2.8%, 7.0%, and 1.4% demonstrating homogeneity of the 2.0, 6.0, and 10.0 mg/mL dose formulations.

Remaining samples were appropriately discarded prior to finalization of the study report.

6. Sample Batch Acceptance Criteria

For a sample batch run to be acceptable, the following assay acceptance criteria were met;

Calibration Curve Acceptance Criteria:

- Accuracy of standards: 20% of nominal (25% near LLOQ)
- Precision of the standards: %CV value $\leq 20.0\%$
- At least 75% of the standards (minimum of 6 points) are required to meet these criteria. Failed standards were removed from the analysis.

QC Sample Acceptance Criteria:

- Accuracy of QC samples: 20% of nominal (25% near LLOQ)
- Precision of the QCs: %CV value $\leq 20.0\%$
- At least 67% of all QC samples are required to meet these criteria. In addition, at least 50% of the QC samples at each level are required to meet these criteria.

Concentration and homogeneity of dosing formulations was determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container. Measured concentration of the dosing formulation was acceptable if it was 80% to 120% of the nominal concentration. Formulated LT3114 was considered homogeneous if the measured concentrations of the top, middle and bottom aliquots had a %CV value $\leq 20.0\%$.

7. Reserve Archive Samples

Since the study was of short duration, no retention sample of the test article was maintained at Calvert.

8. Accountability and Disposition

Unused test article was retained for use in related studies.

B. Test System (Animals and Animal Care)

1. Description

Species:	Rat
Stock:	Hsd:SD

Total Number:	Forty (20/sex): five/sex per test article dose level and five/sex for vehicle controls
Gender:	Male and Females
Age:	6 weeks – males 7 weeks - females
Body Weight Range:	195-213 grams – males 181-197 grams - females
Animal Source:	Harlan Laboratories
Experimental History:	Purpose-bred and experimentally naïve at the outset of the study.
Identification:	Ear tagged and color coded

2. *Rationale for Choice of Species and Number of Animals*

Rats are widely used for the evaluation of neuropharmacological response to test agents (1, 2). Previous research has been conducted with the test article in rats.

Based on available literature and Calvert's experience with this model, the number of animals used in this study is the minimum amount considered necessary to provide valid scientific assessment (3).

3. *Husbandry*

Housing:	Animals were group housed in compliance with the National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals were kept was documented in the study records. No other species were kept in the same room. The experiment was conducted in a room dedicated to this study and the dosing occurred in this room only.
Lighting:	12 hours light/12 hours dark

Room Temperature:	19 to 23°C
Relative Humidity:	38 to 76%
Food:	All animals had access to Certified Rodent Diet (Teklad) equivalent <i>ad libitum</i> , except during the neuropharmacological evaluations. The lot number and specifications of each lot used are archived at Calvert. No contaminants were known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which are maintained in the Calvert archives.
Water:	Water was available <i>ad libitum</i> , except during the neuropharmacological evaluations, to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOP's. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives.
Acclimation:	Study animals were acclimated to their housing for 5 days prior to dosing.

4. *Prestudy Health Screen and Selection Criteria*

All animals received for this study were assessed as to their general health by veterinary staff or other suitably trained authorized personnel. During the acclimation period, each rat was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use were assigned to this study.

5. Assignment to Study Groups

Animals were selected by body weight and apparent good health and were randomly assigned to study groups. Rats were given the following identification numbers and identified by ear tag and color code:

Group Number	Color Code	Males	Females
1	Green	1801-1805	1806-1810
2	Blue	1811-1815	1816-1820
3	Purple	1821-1825	1826-1830
4	Red	1831-1835	1836-1840

6. Humane Care of Animals

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 2011, The National Academies Press). The Calvert IACUC approved the study protocol prior to finalization.

C. Test Article Administration

1. Group Assignments and Dose Levels

Group	Treatment	Dose (mg/kg)	Volume (mL/kg)	Concentration (mg/mL)	Number of Animals per sex	
					Male	Female
1	Diluent	0	5	0	5	5
2	LT3114	10	5	2.0	5	5
3	LT3114	30	5	6	5	5
4	LT3114	50	5	10	5	5

2. Dosing

Route: Slow intravenous injection (~1 minute)

Frequency: Once

Procedure: Doses were administered by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg. Each intravenous dose was administered as a manual slow injection (~1 minute). Each animal

received a mg/kg dose based on its most recent body weight.

3. *Justification for Route and Dose Levels*

The intravenous route was chosen as it is the intended route of administration in humans.

The dose levels were selected by the Sponsor based on in-life study results of a 14-day repeat intravenous dose range finding toxicity study of LT3114 in Sprague Dawley Rats. (Calvert study 0437RL35.002).

D. Method of Study Performance

The rats were weighed, dosed, and placed in a fixed environment consisting of a Plexiglas® enclosure, fitted with a lid. The enclosure was placed on absorbent paper which detects excretions. In this environment, the rats were free to move about. The rats were observed for signs of pharmacological or toxicological activity at 15, 30 and 45 minutes and 1, 2, 3, 4, and 24 hours following treatment with the results recorded. All observations were made at ± 5 -15 minutes of the designated times as appropriate. Observances for the following symptoms were made:

Seizures/convulsions	Awareness reaction
Startle response	Vocalization
Irritability	Decreased abdominal tone
Increased secretion	Body tremors
Decreased grip strength	Immobility
Motor activity	Ataxia
Abnormal posture	Stereotypy
Excretion	Decreased respiration
Piloerection	Loss of righting
Pupil size	Nociceptive (pain) response
Corneal reflex	Pinnal reflex

Body temperatures (rectal) were taken on all animals with a YSI Precision Digital Thermometer Model 4600 at 30 minutes (± 5 minutes) following dose administration.

E. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

All animals were euthanized by CO₂ asphyxiation following the 24 hour observation and carcasses discarded.

VIII. Records and Reports

A. Data Collection/Statistical Analysis

All data was manually generated and recorded on paper.

No statistical analysis was performed for behavioral tests, as the data generated is not quantitative enough to allow for meaningful comparison.

Body temperature values are presented as Mean $\pm$ SEM and statistically significant effects were determined by ANOVA with Tukey HSD Multiple Comparison Test using Systat v.9.01. A p value ≤ 0.05 denotes statistical significance.

B. Storage of Records

Test article preparation, test article tracking, in-life data, protocol, protocol amendment(s) and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

All dose formulation data will be archived at Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560.

IX. Results

A. Neurobehavioral Evaluations

As presented in Tables 1, 2, and 3, no apparent neuropharmacological signs were observed through 24 hours post-dose in any rats receiving the vehicle at 5 mL/kg or LT3114 at doses of 10, 30 or 50 mg/kg by intravenous injection.

Body temperature was not affected by any of the dose levels at 30 minutes post-dose.

X. Conclusion

The intravenous administration of LT3114 at 10, 30 and 50 mg/kg did not produce any apparent neuropharmacological effects in male or female rats. Body temperature was not affected by the intravenous administration of LT3114 at doses up to 50 mg/kg.

XI. References

1. S. Irwin, "Drug screening and evaluation of new drugs in animals," in *Animal and Clinical Pharmacologic Techniques in Drug Evaluation* J. M. Nodine and P. E. Siegler, eds.(Year Book Medical Publishers, Inc., (1964): 36-54.
2. S. Irwin, "Comprehensive Observational Assessment: Ia. A Systematic, Quantitative Procedure for Assessing the Behavioral and Physiologic State of the Mouse". *Psychopharmacologia (Berl)*, 13: 222-257 (1968).
3. Y. Zhu, M.R. Wessel, T. Liu, V.C. Moser, "Analyses of Neurobehavioral Screening Data: Dose-Time-Response Modeling of Continuous Outcomes." *Regulatory Toxicology and Pharmacology*, 41 (3): 240-255 (2005).

XII. Tables

A. Table 1 - Mean Body Temperatures and Observed Signs (n=10, 5/sex)

Group No.	Intravenous Treatment	Dose (mg/kg)	Volume (mL/kg)	Sex	30 min pd Body Temp (°C) ^a	Signs Observed (15 min-24 hr post-dose)
1	Vehicle (LT3114 Diluent)	0	5	Male	38.6 ± 0.17	No signs
				Female	39.3 ± 0.08	
2	LT3114	10	5	Male	38.9 ± 0.10	No signs
				Female	39.3 ± 0.02	
3	LT3114	30	5	Male	38.9 ± 0.18	No signs
				Female	39.1 ± 0.07	
4	LT3114	50	5	Male	38.7 ± 0.15	No signs
				Female	39.1 ± 0.17	

pd = post-dose

^a Data are presented as mean ± standard error of the mean^b There were no statistically significant ($p \leq 0.05$) changes in body temperature compared to the vehicle control group – ANOVA followed by a Tukey HSD Multiple Comparison Test.

B. Table 2 - Individual Animal Data

Group Number	Animal Number	Sex	Intravenous Treatment	30 Minute Post-dose Body Temperature (°C)
1	1801	Male	Vehicle (LT3114 Diluent)) 0 mg/kg 5 mL/kg 0 mg/mL	38.26
	1802	Male		39.25
	1803	Male		38.52
	1804	Male		38.46
	1805	Male		38.71
	1806	Female		39.32
	1807	Female		39.11
	1808	Female		39.17
	1809	Female		39.45
	1810	Female		39.51
2	1811	Male	LT3114 10 mg/kg 5 mL/kg 2.0 mg/mL	38.73
	1812	Male		38.90
	1813	Male		38.83
	1814	Male		38.86
	1815	Male		39.33
	1816	Female		39.23
	1817	Female		39.33
	1818	Female		39.34
	1819	Female		39.30
	1820	Female		39.33

B. Table 2 (continued) - Individual Animal Data

Group Number	Animal Number	Sex	Oral Treatment	30 Minute Post-dose Body Temperature (°C)
3	1821	Male	LT3114 30 mg/kg 5 mL/kg 6 mg/mL	39.22
	1822	Male		38.77
	1823	Male		38.84
	1824	Male		38.41
	1825	Male		39.47
	1826	Female		38.87
	1827	Female		39.05
	1828	Female		38.96
	1829	Female		39.27
	1830	Female		39.17
4	1831	Male	LT3114 50 mg/kg 5 mL/kg 10 mg/mL	38.31
	1832	Male		38.51
	1833	Male		39.12
	1834	Male		38.94
	1835	Male		38.86
	1836	Female		38.97
	1837	Female		39.08
	1838	Female		38.67
	1839	Female		39.71
	1840	Female		39.09

C. Table 3 – Group Neuropharmacological or Toxicological Signs

Group Number: 1 (n=10, 5/sex)

Dose: Vehicle (5 mL/kg)

Animal Numbers: 1801-1810

Time of Observations

Signs	15 min	30 min	45 min	1 hr	2 hr	3 hr	4 hr	24 hr
Seizures/convulsions	-	-	-	-	-	-	-	-
Awareness reaction	-	-	-	-	-	-	-	-
Startle response	-	-	-	-	-	-	-	-
Vocalization	-	-	-	-	-	-	-	-
Irritability	-	-	-	-	-	-	-	-
Decreased Abdominal tone	-	-	-	-	-	-	-	-
Increased Secretion	-	-	-	-	-	-	-	-
Body tremors	-	-	-	-	-	-	-	-
Decreased Grip strength	-	-	-	-	-	-	-	-
Immobility	-	-	-	-	-	-	-	-
Motor activity	-	-	-	-	-	-	-	-
Ataxia	-	-	-	-	-	-	-	-
Abnormal Posture	-	-	-	-	-	-	-	-
Stereotypy	-	-	-	-	-	-	-	-
Excretion	-	-	-	-	-	-	-	-
Decreased Respiration	-	-	-	-	-	-	-	-
Piloerection	-	-	-	-	-	-	-	-
Loss of righting	-	-	-	-	-	-	-	-
Pupil size	-	-	-	-	-	-	-	-
Nociceptive (pain) response	-	-	-	-	-	-	-	-
Corneal reflex	-	-	-	-	-	-	-	-
Pinnal reflex	-	-	-	-	-	-	-	-

- No signs

C. Table 3 (continued) – Group Neuropharmacological or Toxicological Signs

Group Number: 2 (n=10, 5/sex)

Dose: LT3114 at 10 mg/kg

Animal Numbers: 1811-1820

Time of Observations

Signs	15 min	30 min	45 min	1 hr	2 hr	3 hr	4 hr	24 hr
Seizures/convulsions	-	-	-	-	-	-	-	-
Awareness reaction	-	-	-	-	-	-	-	-
Startle response	-	-	-	-	-	-	-	-
Vocalization	-	-	-	-	-	-	-	-
Irritability	-	-	-	-	-	-	-	-
Decreased Abdominal tone	-	-	-	-	-	-	-	-
Increased Secretion	-	-	-	-	-	-	-	-
Body tremors	-	-	-	-	-	-	-	-
Decreased Grip strength	-	-	-	-	-	-	-	-
Immobility	-	-	-	-	-	-	-	-
Motor activity	-	-	-	-	-	-	-	-
Ataxia	-	-	-	-	-	-	-	-
Abnormal Posture	-	-	-	-	-	-	-	-
Stereotypy	-	-	-	-	-	-	-	-
Excretion	-	-	-	-	-	-	-	-
Decreased Respiration	-	-	-	-	-	-	-	-
Piloerection	-	-	-	-	-	-	-	-
Loss of righting	-	-	-	-	-	-	-	-
Pupil size	-	-	-	-	-	-	-	-
Nociceptive (pain) response	-	-	-	-	-	-	-	-
Corneal reflex	-	-	-	-	-	-	-	-
Pinnal reflex	-	-	-	-	-	-	-	-

- No signs

C. Table 3 (continued) – Group Neuropharmacological or Toxicological Signs

Group Number: 3 (n=10, 5/sex)

Dose: LT3114 at 30 mg/kg

Animal Numbers: 1821-1830

Time of Observations

Signs	15 min	30 min	45 min	1 hr	2 hr	3 hr	4 hr	24 hr
Seizures/convulsions	-	-	-	-	-	-	-	-
Awareness reaction	-	-	-	-	-	-	-	-
Startle response	-	-	-	-	-	-	-	-
Vocalization	-	-	-	-	-	-	-	-
Irritability	-	-	-	-	-	-	-	-
Decreased Abdominal tone	-	-	-	-	-	-	-	-
Increased Secretion	-	-	-	-	-	-	-	-
Body tremors	-	-	-	-	-	-	-	-
Decreased Grip strength	-	-	-	-	-	-	-	-
Immobility	-	-	-	-	-	-	-	-
Motor activity	-	-	-	-	-	-	-	-
Ataxia	-	-	-	-	-	-	-	-
Abnormal Posture	-	-	-	-	-	-	-	-
Stereotypy	-	-	-	-	-	-	-	-
Excretion	-	-	-	-	-	-	-	-
Decreased Respiration	-	-	-	-	-	-	-	-
Piloerection	-	-	-	-	-	-	-	-
Loss of righting	-	-	-	-	-	-	-	-
Pupil size	-	-	-	-	-	-	-	-
Nociceptive (pain) response	-	-	-	-	-	-	-	-
Corneal reflex	-	-	-	-	-	-	-	-
Pinnal reflex	-	-	-	-	-	-	-	-

- No signs

C. Table 3 (continued) – Group Neuropharmacological or Toxicological Signs

Group Number: 4 (n=10, 5/sex)

Dose: LT3114 at 50 mg/kg

Animal Numbers: 1831-1840

Time of Observations

Signs	15 min	30 min	45 min	1 hr	2 hr	3 hr	4 hr	24 hr
Seizures/convulsions	-	-	-	-	-	-	-	-
Awareness reaction	-	-	-	-	-	-	-	-
Startle response	-	-	-	-	-	-	-	-
Vocalization	-	-	-	-	-	-	-	-
Irritability	-	-	-	-	-	-	-	-
Decreased Abdominal tone	-	-	-	-	-	-	-	-
Increased Secretion	-	-	-	-	-	-	-	-
Body tremors	-	-	-	-	-	-	-	-
Decreased Grip strength	-	-	-	-	-	-	-	-
Immobility	-	-	-	-	-	-	-	-
Motor activity	-	-	-	-	-	-	-	-
Ataxia	-	-	-	-	-	-	-	-
Abnormal Posture	-	-	-	-	-	-	-	-
Stereotypy	-	-	-	-	-	-	-	-
Excretion	-	-	-	-	-	-	-	-
Decreased Respiration	-	-	-	-	-	-	-	-
Piloerection	-	-	-	-	-	-	-	-
Loss of righting	-	-	-	-	-	-	-	-
Pupil size	-	-	-	-	-	-	-	-
Nociceptive (pain) response	-	-	-	-	-	-	-	-
Corneal reflex	-	-	-	-	-	-	-	-
Pinnal reflex	-	-	-	-	-	-	-	-

- No signs

XIII. Appendices

A. Appendix I - Test Article Information

Lpath Inc.	Certificate of Analysis
-------------------	--------------------------------

TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing	REVISION No.
		1.0.0
		PAGE 1 of 2

Name		Part Number		Lot Number
Lpathomab-DS, 30 mg/mL		TBD		GA121459
Formulation Buffer				Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0				2 to 8°C
Production Date		Test Date	Retest Date	Intended Use
JUN 2014		JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer		Manufacturer Part Number		Manufacturer Lot Number
Gallus-MO, St. Louis, MO		TBD		Gallus-MO 103948
Attribute	Test	Specification		Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution		Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2		5.1
	CGE (Non-reduced)	Major band at 150 kDa		Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa		Conforms
	iCE	Comparable to RS		Conforms
		Minor Peak at pI ~8.3		15%
Major Peak at pI ~8.4		60%		
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL		32.1 mg/mL
		100% ±40% of RS		89%
	LPA Binding			

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

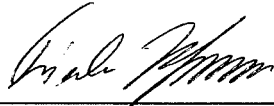
Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance:


Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

Lpath Certificate of Analysis																
Stability Assessment of Lpathomab-DS, lot 103948, stored at 5 ± 3 °C					Formulation Age											
Assay		Specification			Time 0		1 Month		2 Months		3 Months					
Appearance		Particulate-free, clear, colorless to slightly yellowish and slightly opalescent solution.			Conforms		Conforms		Conforms		Conforms					
pH		4.8 to 5.2			5.2		5.1		5.1		5.1					
ICE		Comparable to Reference Standard			Conforms		Conforms		Conforms		Conforms					
		Report pI and relative peak area of Major Isoform and total relative peak area of acidic and basic variants			pI		Area		pI		Area		pI		Area	
					Acidic		23%		Acidic		22%		Acidic		23%	
					8.2		53%		8.2		53%		8.2		53%	
					Basic		24%		Basic		25%		Basic		24%	
CGE-NR		IgG + IgG-NG ≥ 90%			98.4%		97.9%		98.8%		97.6%					
		IgG-NG			2.1%		1.8%		1.5%		2.2%					
		2H-1L			0.3%		0.4%		0.4%		0.4%					
		2H			0%		0%		0%		0%					
		HL			0%		0%		0%		0%					
		HC			0%		0%		0%		0%					
		LC			0.4%		0.4%		0.5%		0.6%					
CGE-R		HC + HC-NG + LC ≥ 90%			98.1%		95.5%		99.1%		98.1%					
		HC			66%		63%		67%		66%					
		HC-NG			1.1%		1.5%		1.1%		1.1%					
		LC			31%		31%		31%		31%					
qSE-HPLC		API concentration: 29 to 35 mg/mL			33.2		33.5%		34.3		34.5					
		Purity: ≥ 90%			99.2		98.6%		98.3%		98.2					
		Aggregates: ≥ 5%			0.8%		1.4%		1.7%		1.8					
		No single impurity > %			0%		0%		0%		0%					
LPA-binding ELISA		60% to 140% of reference standard activity			125%		106%		92%		136%					
Particulate		≤ 6,000 Particles ≥ 10 µm per vial			51						51					
		≤ 600 Particles ≥ 25 µm per vial			2						0					
Judgment																
Lot 103948 of Lpathomab-DS was placed in a validated 5 ± 3 °C stability chamber at Lpath's qualified contract laboratory, Eurofins Lancaster Laboratories in Lancaster, PA, and tested monthly for three months with the stability-indicating assays appearance, pH, isoelectric focusing by capillary electrophoresis (ICE), non-reduced and reduced capillary gel electrophoresis (CGE-NR and CGE-R), quantitative size-exclusion high performance liquid chromatography (qSE-HPLC), LPA-binding ELISA and subvisible particulate matter. The obtained results and underlying raw data were reviewed and verified by Lpath's Quality Department. The results only varied within each test-specific accuracy tolerance and indicate that the lot exhibited excellent product stability under the storage conditions.																
Department		Title		Name			Signature			Date						
Quality Assurance		Head of Product Quality		Frieder K Hofmann, PhD						22DEC2014						

B. Appendix II - Test Article Formulation Analyses

STUDY PHASE REPORT

STUDY TITLE: Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

REPORT TITLE: Determination of LT3114 in Dose Formulation Samples

BRT Reference No.: BRT 20140813

Author: Victor J. Johnson, PhD (BRT)

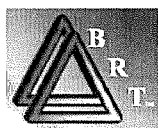
**Dosing Formulation
Analyses Laboratory:** Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Number: 0200RL35.001

Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121-1404

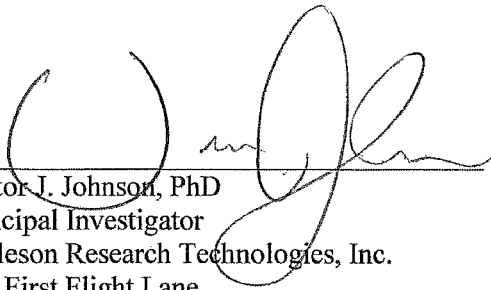
Report Date: 19 December 2014



COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)
- OECD Principles on Good Laboratory Practice (C[97]186/Final)

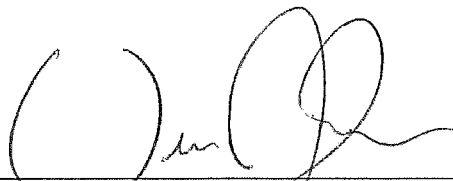


Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
USA

19 Dec 14

Date

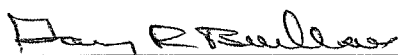
APPROVALS



Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.

19 Dec 14

Date

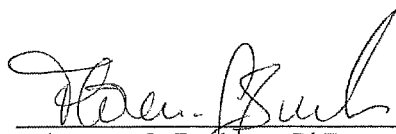


Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

19 Dec 2014

Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

19 Dec 14

Date

QUALITY ASSURANCE STATEMENT

Study Phase Reference Number: BRT 20140813

Study Phase Title: Determination of LT3114 in Dose Formulation Samples

Testing Site: Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560

Principal Investigator: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to BRT Principal Investigator/Management	Date Report Sent to Calvert Study Director/Management
Sample Preparation Audit	09 Dec 14	09 Dec 14	09 Dec 14
Raw Data Audit	17 Dec 14	17 Dec 14	19 Dec 14
Report Audit	17 Dec 14	17 Dec 14	19 Dec 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.


Christine Ford
Quality Assurance Manager

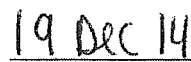

Date

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STUDY INFORMATION

Determination of LT3114 in Dose Formulation Samples

Study Title:	Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)
Testing Facility:	Calvert Laboratories, Inc.
Testing Facility Study No:	0200RL35.001
Study Director:	Brian Klatt, B.S. (Calvert Laboratories)
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121-1404
Test Site (Dosing Formulation Analyses):	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Test Site Reference No.:	BRT 20140813
Principal Investigator:	Victor J. Johnson, PhD (BRT)
Analyst:	Kristen Fisher (BRT)
Study Initiation Date:	01 Aug 14
Sample Receipt Date:	04 Sep 14
Experimental Start Date:	03 Dec 14
Experimental Completion Date:	12 Dec 14
Study Phase Completion Date:	19 Dec 14

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples generated by Calvert Study No. 0200RL35.001:

“Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)”.

Dose formulation samples were provided for three dose concentrations, 2.0 mg/mL, 6.0 mg/mL, and 10.0 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle, from the top, middle, and bottom strata of the LT3114 dose formulations and from the middle stratum of the vehicle. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.0 – 2.1 mg/mL, 5.7 – 6.5 mg/mL, and 10.6 – 10.9 mg/mL for the 2.0, 6.0, and 10.0 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision. LT3114 concentrations from the top, middle, and bottom fractions of the dose formulations had a CV% of 2.8%, 7.0%, and 1.4% demonstrating homogeneity of the 2.0, 6.0, and 10.0 mg/mL dose formulations.

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples in support of the study 0200RL35.001 entitled “Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)”.

STUDY DESIGN

Dose Formulation Samples

Dose formulations (2.0, 6.0, and 10.0 mg/mL) were prepared by Calvert Laboratories on 11 Aug 14 according to the study protocol, using LT3114 in a citrate buffer formulation supplied by the Sponsor. Duplicate 1 mL samples were collected on 11 Aug 14 from the top, middle, and bottom of each dosing solution and from the middle of the vehicle solution. The samples were stored at 2-8°C until single aliquots from each sample were shipped to BRT and received on 04 Sep 14 (Table 1). Dose formulation samples were logged into BRT inventory and stored at 2-8°C until analyzed for LT3114 concentration. Samples were returned to 2-8°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director per protocol. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received at BRT on 04 Sep 14.

Date of Collection	Date of Receipt at BRT	Aliquots	LT3114 Concentration (mg/mL)	Location	Volume (mL)
11 Aug 14	04 Sep 14	1	Vehicle	Middle	1
11 Aug 14	04 Sep 14	1	2.0	Top	1
11 Aug 14	04 Sep 14	1	2.0	Middle	1
11 Aug 14	04 Sep 14	1	2.0	Bottom	1
11 Aug 14	04 Sep 14	1	6.0	Top	1
11 Aug 14	04 Sep 14	1	6.0	Middle	1
11 Aug 14	04 Sep 14	1	6.0	Bottom	1
11 Aug 14	04 Sep 14	1	10.0	Top	1
11 Aug 14	04 Sep 14	1	10.0	Middle	1
11 Aug 14	04 Sep 14	1	10.0	Bottom	1

METHODOLOGY

Dose Formulation Samples

Dose formulation samples (single Vehicle sample plus 3 dose levels with 3 samples each from top, middle, and bottom of the preparations = 10 total samples) were received in

good condition on wet ice on 04 Sep 14. Upon receipt at BRT, all samples were stored at 2-8°C. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples were analyzed for LT3114 concentration using a validated ELISA method (BRT 20140629A). The ELISA method is provided in Appendix 1.

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, delipidized bovine serum (DBS) assay buffer, and dose formulation samples were added to the coated plates and incubated to allow for capture of the LT3114. A horseradish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data and Statistical Analysis

Data analysis was performed using SoftMax[®]Pro (Molecular Devices) software. Calibration curves were generated by plotting the mean OD₄₅₀ values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD₄₅₀ values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from duplicate wells, by the sample dilution factor. Results were reported to one decimal place.

Calculations, including means, standard deviations, and coefficients of variation, for determining accuracy and precision were performed using Microsoft Excel 2013.

Acceptance Criteria (per method validation BRT 20140629A)

Calibration Curve Acceptance Criteria:

- Calibration curve range: 3.9 to 500 ng/mL
- Accuracy of standards: 20% of nominal (25% at LLOQ and ULOQ)
- Precision of the standards within 20%

- At least 75% of the standards (minimum of 6 points) met these criteria. Failed standards were removed from the analysis, if present.

QC Sample Acceptance Criteria:

- QC1-300 ng/mL; QC2-150 ng/mL; QC3-5 ng/mL
- Accuracy of QC samples: 20% of nominal (25% at 5 ng/mL and 300 ng/mL)
- Precision of the QCs were within 20% (25% at 5 ng/mL and 300 ng/mL)
- At least 67% of all QC samples met these criteria. In addition, at least 50% of the QC samples at each level met these criteria.
- Total error ($|\%RE| + \%CV$) of each QC was $\leq 30\%$ ($\leq 40\%$ at 5 ng/mL and 300 ng/mL)

Study Dose Formulation Sample Acceptance Criteria:

- Precision of the replicates was within 20% (25% at LLOQ and ULOQ)
- Sample concentrations were acceptable when accuracy was within 20% of nominal

Study Dose Formulation Sample Reanalysis:

- Samples were reanalyzed if the assay is rejected due to failure of the standards or QCs to meet acceptance criteria provided above.
- Samples were reanalyzed if the mean OD450 for the replicates is outside the range of OD450 values for the calibration curve. If applicable, the sample was diluted appropriately and reanalyzed. This did not apply to samples diluted at the MRD that were below the range of the standard curve.
- For samples that were within the range of the standard curve, samples were reanalyzed if the precision of the replicates was not acceptable.

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples (top, middle, and bottom) of each dose formulation are provided in Table 2. Analysis confirmed the absence of LT3114 in the vehicle preparations. The concentrations of LT3114 in the 2.0 mg/mL dose formulations were 2.0, 2.1, and 2.1 mg/mL for the 2.0 mg/mL dose formulation, 5.7, 6.4, and 6.5 mg/mL for the 6.0 mg/mL dose formulation, and 10.6, 10.8, and 10.9 mg/mL for the 10.0 mg/mL

dose formulation. All dose formulation concentration met acceptance criteria for accuracy and precision. The homogeneity of each dose formulation was assessed by determining the CV% for the top, middle, and bottom samples collected from each formulation. The CV% was 2.8%, 7.0%, and 1.4% for the 2.0, 6.0, and 10.0 mg/mL dose formulations, respectively, demonstrating acceptable homogeneity for all dose formulations.

All dose formulation samples were analyzed within the 2-8°C stability for LT3114 in a citrate buffer formulation established in the method validation (BRT 20140629A).

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 2 years from the date that the final study phase report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

- There was a technical error in preparation of the calibration curve for the assay run performed on 09 Dec 14 that resulted in failure of the QC samples to meet acceptance criteria. The assay run was rejected and the analysis was repeated. There was no impact of these deviation on the study phase since the analysis was successfully repeated.

TABLES

Table 1: Dose formulation samples received at BRT on 04 Sep 14.	8
Table 2: LT3114 concentrations in dose formulation samples.	13

Table 2: LT3114 concentrations in dose formulation samples.

Sampling Dates	Dose Formulation	Location	LT3114 (mg/mL)	% of Target	CV%
11 Aug 14	Vehicle	Middle	BLQ	N/A	N/A
	LT3114 Low-Dose: 2.0 mg/ml	Top	2.1	105	2.8
	LT3114 Low-Dose: 2.0 mg/ml	Middle	2.1	105	
	LT3114 Low-Dose: 2.0 mg/ml	Bottom	2.0	100	
	LT3114 Mid-Dose: 6.0 mg/ml	Top	5.7	95	7.0
	LT3114 Mid-Dose: 6.0 mg/ml	Middle	6.4	107	
	LT3114 Mid-Dose: 6.0 mg/ml	Bottom	6.5	108	
	LT3114 High-Dose: 10.0 mg/ml	Top	10.9	109	1.4
	LT3114 High-Dose: 10.0 mg/ml	Middle	10.8	108	
	LT3114 High-Dose: 10.0 mg/ml	Bottom	10.6	106	

BLQ – Below Lower Limit of Quantification (3.9 ng/mL); N/A – Not applicable.

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material was kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material was stored at 2-8° C for up to 4 months.
2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA l, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 300, 150, and 10 ng/mL. Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 2.5 $\mu\text{g/ml}$ in 0.05 M Carbonate Buffer.
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 μL /well coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.

- 3.2 Dispense 150 µL/well of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour ± 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 8 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.

8. Stop Solution Addition

8.1 Dispense 100 μ l 1.0 M H₂SO₄ solution to each well.

8.2 Measure absorbance at 450 nm on microplate reader.

C. Appendix III – Protocol, Protocol Amendments and Deviations



Study Protocol

Title: Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

Calvert Study No.: 0200RL35.001

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

LABORATORY

Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447 USA
Phone: 570 566 2411
Fax: 570 566 3450

www.calvertlabs.com

CORPORATE

1225 Crescent Green Dr
Suite 115
Cary, NC 27518
Phone: 919 854 4453
Fax: 919 854 2860

www.calvertholdings.com

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II. Introduction

A. Title

Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)

B. Objective

The purpose of this study is to evaluate the potential neuropharmacological effects of LT3114 in male and female rats.

C. Regulatory Compliance

This Study will be conducted in compliance with the Good Laboratory Practice Regulations 21 CFR Part 58, Japanese Ministry of Health, Labour and Welfare Ordinance 21 issued March 26, 1997, effective April 1, 1997, in accordance with OECD Principles of Good Laboratory Practice as revised 26 November 1997 [C(97)186/final], and in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert.

Study elements conducted by subcontractors will be performed in accordance with the appropriate Standard Operating Procedures of the subcontractor.

D. Testing Guidelines

FDA Guidance for Industry "S7A Safety Pharmacology Studies for Human Pharmaceuticals (ICH)" issued July 2001

E. Declaration of Intent

This study may be submitted to the United States Food and Drug Administration (FDA), an Organization for Economic Cooperation and Development (OECD) member country, and/or the Japanese Ministry of Health, Labor and Welfare (JMHLW).

F. Calvert Study Number

0200RL35.001

G. Testing Facility

Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

H. Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121-1404

I. Study Director

Brian Klatt, B.S.
Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447
Phone: (570) 585-2229
Fax: (570) 585-2383
E-mail: brian.klatt@calvertlabs.com

J. Study Monitor

Marlene W. Modi, Ph.D.
MWM Consulting Group, Inc. for Lpath Inc.
14 Joseph Court
Monmouth Junction, NJ 08852
Phone: (732) 274-9562
Mobile: (732) 822-6101
E-mail: mwmconsultinggroup@gmail.com

K. Sponsor Representative

Gary Woodnutt, PhD
Senior Vice President, Research
Lpath Inc
4025 Sorrento Valley Blvd

San Diego, CA 92121
Phone: (858) 678-0800 ext 112
Email: rmatteo@lpath.com

L. Principal Investigator for Dosing Formulation Analyses

Victor T. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

M. Lead Quality Assurance - Calvert

Darlene Gilpin, RQAP-GLP
Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447
Phone: (570) 585-2219
Fax: (570) 585-2383
E-mail: darlene.gilpin@calvertlabs.com

N. Key Study Dates

Proposed Study Initiation Date:	1 Aug 2014
Proposed First Day of Dosing:	13 Aug 2014
Proposed Experimental Completion Date:	14 Aug 2014

III. Materials and Methods

A. Test Article

1. Test Article

Identification: LT3114

Lot/Batch No.: 103948

Physical Description: Clear colorless liquid

Storage Conditions: Stored refrigerated (2-8°C) and protected from light.

Characterization/
Stability: Good Laboratory Practice (GLP) regulations require that all test articles be appropriately characterized for identity, purity, composition and stability. Compliance necessitates documentation that characterization and stability testing have been performed. The sponsor will provide the Study Director with a certificate of analysis/results of stability testing and whether analysis/testing was conducted under GLP/GMP. If characterization/stability is not provided or was not performed under GLP/GMP, this fact will be reflected in the compliance statement.

2. Vehicle

Identification: LT3114 Diluent

Lot/Batch No.: 104011-001

Physical Description: Clear colorless liquid

Storage Conditions: 2-8°C

3. Dose Preparation

The Sponsor will provide the test article already formulated at 32.1 mg/ml in sterile containers. Diluent will also be provided in separate sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared.

Do not shake the formulation. Group 1 control animals will receive the diluent. When necessary depending on the dose level, doses will be prepared by dilution using the provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dose preparation, 2 mL of the 10% Polysorbate 80 solution provided, should be added aseptically to each liter of formulation buffer, also supplied, and mixed gently to ensure distribution throughout the volume of buffer. This diluent will then be used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

4. Dose Formulation Sampling

On the day of dosing, duplicate 1.0 mL samples of each test article formulation will be collected from the top, middle and bottom of each formulation to confirm the concentration and homogeneity. Duplicate samples (1.0 mL) from the middle of the vehicle will be collected to confirm the absence of test article. These samples will be stored at 2-8°C until shipment for dose formulation analysis.

5. Shipment and Analysis

Dose formulation samples will be shipped to (on dry ice) and analyses will be performed under the direction of:

Victor T. Johnson, Ph.D.
Grants & Contracts PI/Study Director
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: vjohnson@brt-labs.com

Good Laboratory Practice (GLP) regulations require that all formulated test articles be appropriately verified for concentration, homogeneity and stability.

Targeted Total Number of

Formulation Samples: 20 Collected, 10 (initial shipment)

Sample Storage:	
Conditions:	Refrigerated (2-8°C)
Sample Ship Date:	Upon approval of analytical laboratory
Testing Requirements:	Concentration and homogeneity
Analytical Methods:	To be added by protocol amendment
Automated Data	
Collection Systems:	To be added by protocol amendment

If any samples are re-analyzed, the Study Director will be notified and details will be included in the analytical report.

All Quality Assurance Unit activities and Quality Assurance Unit reports for the bioanalytical analyses will be sent to the Calvert Study Director and Calvert Management. All protocol, SOP and GLP deviations associated with the bioanalytical analyses will be reported to the Calvert Study Director. The final bioanalytical report will have a GLP compliance statement and a Quality Assurance Unit statement which will have the dates of any inspections and the dates the QAU reports were sent to the Principal Investigator, Study Director, and Study Director's Management.

A report of the sample analysis procedures and results will be provided to the Study Director for inclusion as an appendix in the study report.

Remaining samples will be appropriately discarded prior to finalization of the study report.

6. Sample Batch Acceptance Criteria

For a sample batch run to be acceptable, a minimum of 75% of the QC samples for LT3114 at each concentration must be within $\pm 10.0\%$ from the nominal value. In addition, a minimum of 75% of the back-calculated CS values for LT-3114 must fall within 10.0% of their nominal values.

Concentration and homogeneity of dosing formulations will be determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container. Measured concentration of the

dosing formulation will be acceptable if it is 85% to 115% of the nominal concentration. Formulated LT3114 will be considered homogeneous if the measured concentrations of the top, middle and bottom aliquots have a %CV value $\leq 10.0\%$.

In the event that a sample is outside of the acceptable specification range, the Study Director will justify the acceptability of the results or suggest re-analysis of the backup samples or retest the affected portion of the study.

7. Reserve Archive Samples

Since the study is of short duration, no retention sample of the test article will be maintained at Calvert.

8. Accountability and Disposition

Unused test article will be returned to the Sponsor designee at study completion (prior to issue of final report) or, if necessary, retained for use on related future studies. The Sponsor will be notified in advance of shipping and a transmittal letter will accompany the shipment. The material will be packed in a suitable container to maintain the conditions specified by the Sponsor during transit plus an adequate margin of safety to account for any possible transit delays.

B. Test System (Animals and Animal Care)

1. Description

Species:	Rat
Strain/Substrain:	CrI: CD®(SD)
Total Number:	Forty (20/sex): five/sex per test article dose level and five/sex for vehicle controls
Gender:	Male and Females
Age Range:	5 to 8 weeks – males 7 to 10 weeks - females
Body Weight Range:	180-250 grams
Animal Source:	Charles River Laboratories or any other approved supplier

Experimental History: Purpose-bred and experimentally naïve at the outset of the study.

Identification: Ear tagged and color coded

2. Rationale for Choice of Species and Number of Animals

Rats are widely used for the evaluation of neuropharmacological response to test agents (1, 2). Previous research has been conducted with the test article in rats.

Based on available literature and Calvert's experience with this model, the number of animals used in this study is the minimum amount considered necessary to provide valid scientific assessment (3).

3. Husbandry

Housing: Animals will be group housed upon receipt in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals will be kept will be documented in the study records. No other species will be kept in the same room. The experiment will be conducted in a room dedicated to this study and the dosing will occur in this room only.

Lighting: 12 hours light/12 hours dark

Room Temperature: 20 to 26°C

Relative Humidity: 30-70%

Food: All animals will have access to Certified Rodent Diet (TEKLAD) or equivalent *ad libitum*, unless otherwise specified. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that

performed by the manufacturer, records of which will be maintained in the Calvert archives.

Water: Water will be available *ad libitum*, to each animal via an automatic watering device unless otherwise specified. The water is routinely analyzed for contaminants as per Calvert's SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis will be maintained in the Calvert archives.

Acclimation: Study animals will be acclimated to their housing for a minimum of 5 days prior to their first day of dosing.

4. *Pre-study Health Screen and Selection Criteria*

All animals received for this study will be assessed as to their general health by a member of the veterinary staff or other suitably trained authorized personnel. During the acclimation period, each animal will be observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

5. *Assignment to Study Groups*

Animals will be selected by body weight and apparent good health and will be randomly assigned to study groups.

6. *Humane Care of Animals*

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge, this study is not unnecessary or duplicative.

C. Test Article Administration

1. Group Assignments and Dose Levels

Group	Treatment	Dose (mg/kg)	Volume (mL/kg)	Concentration (mg/mL)	Number of Animals per sex	
					Male	Female
1	Diluent	0	5	0	5	5
2	LT3114	10	5	2.0	5	5
3	LT3114	30	5	6	5	5
4	LT3114	50	5	10	5	5

2. Dosing

Route: Slow intravenous injection (~1 minute)

Frequency: Once

Procedure: Doses will be administered by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg. Each intravenous dose will be administered as a manual slow injection (~1 minute). Each animal will receive a mg/kg dose based on its most recent body weight.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans.

The dose levels were selected by the Sponsor based on in-life study results of a 14-day repeat intravenous dose range finding toxicity study of LT3114 in Sprague Dawley Rats. (Calvert study 0437RL35.002).

D. Method of Study Performance

The rats will be weighed, dosed, and placed in a fixed environment consisting of a Plexiglas® enclosure, fitted with a lid. The enclosure will be placed on absorbent paper which detects excretions. In this environment, the rats will be free to move about. The rats will be observed for signs of pharmacological or toxicological activity at 15, 30 and 45 minutes and 1, 2, 3, 4, and 24 hours following treatment with the results recorded. All observations will be made at ± 5 -15 minutes of the designated times as appropriate. Observances for the following symptoms will be made:

Seizures/convulsions	Awareness reaction
Startle response	Vocalization
Irritability	Decreased abdominal tone
Increased secretion	Body tremors
Decreased grip strength	Immobility
Motor activity	Ataxia
Abnormal posture	Stereotypy
Excretion	Decreased respiration
Piloerection	Loss of righting
Pupil size	Nociceptive (pain) response
Corneal reflex	Pinnal reflex

Body temperatures will be taken on all animals at 30 minutes (± 5 minutes) following dose administration.

E. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

All surviving animals will be euthanized by CO₂ asphyxiation following the 24 hour observation. No necropsies will be performed, no tissues will be saved, and the carcasses will be discarded.

b) Unscheduled Sacrifice

Animals may be euthanized by CO₂ asphyxiation for humane reasons based upon criteria outlined in Calvert's SOP VET-14, "Criteria for Determination of Moribund Animals and for Sacrifice for Humane Reasons"

IV. Records and Reports

A. Data Collection and Statistical Analysis

All study data will be manually generated and recorded on paper.

No statistical analysis will be performed for behavioral tests, as the data generated is not quantitative enough to allow for meaningful comparison.

Body temperature values will be presented as Mean $\pm$ SEM and statistically significant effects will be determined by ANOVA with Tukey HSD Multiple Comparison Test (Systat V. 9.01). A p value ≤ 0.05 will denote statistical significance.

B. Storage of Records

Test article preparation, test article tracking, in-life data, protocol, protocol amendments (if applicable), and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

All dose formulation data will be archived at Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560.

Six months following the submission of the audited draft report if there are no client comments generated by the Sponsor and/or Sponsor Representative, the Sponsor/Sponsor Representative will be notified and the report will be finalized and archived according to the terms stated in the protocol.

V. Miscellaneous

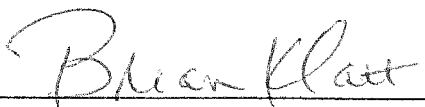
A. Confidentiality Statement

The information contained herein is for the exclusive use of the intended recipient(s).

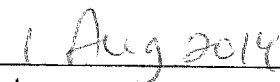
B. References

1. S. Irwin, "Drug screening and evaluation of new drugs in animals," in *Animal and Clinical Pharmacologic Techniques in Drug Evaluation*, J. M. Nodine and P. E. Siegler, eds. (Year Book Medical Publishers, Inc., (1964): 36-54.
2. S. Irwin, "Comprehensive Observational Assessment: Ia. A Systematic, Quantitative Procedure for Assessing the Behavioral and Physiologic State of the Mouse". *Psychopharmacologia (Berl)*, 13: 222-257 (1968).
3. Y. Zhu, M.R. Wessel, T. Liu, V.C. Moser, "Analyses of Neurobehavioral Screening Data: Dose-Time-Response Modeling of Continuous Outcomes." *Regulatory Toxicology and Pharmacology*, 41 (3): 240-255 (2005).

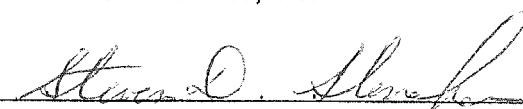
VI. Protocol Approval Signatures



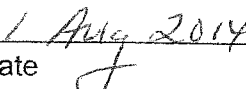
Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.



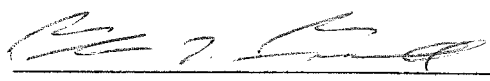
Date



Scientific Management
Calvert Laboratories, Inc.



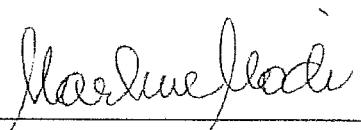
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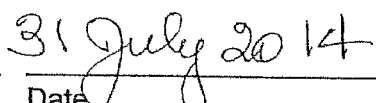
IACUC
Calvert Laboratories, Inc.



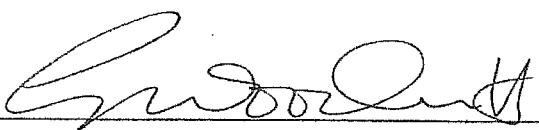
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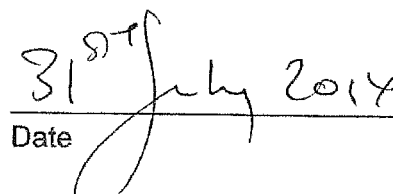
Marlene Modi, Ph.D.
Study Monitor



Date



Gary Woodnutt, Ph.D.
Sponsor Representative



Date



PROTOCOL AMENDMENT

STUDY NUMBER: 0200RL35.001

STUDY TITLE: Neuropharmacological Profile (NPP) of LT3114 in Rats
(GLP)

AMENDMENT NUMBER: 1

EFFECTIVE DATE: 14 Aug 2014

PROTOCOL SECTION CHANGE (Page 4 of 17):

II. Introduction

C. Regulatory Compliance

This study will be conducted in compliance with the Good Laboratory Practice Regulations 21 CFR Part 58, ~~Japanese Ministry of Health, Labour and Welfare Ordinance 21 issued March 26, 1997, effective April 1, 1997,~~ **and** in accordance with OECD Principles of Good Laboratory Practice as revised on 26th November, 1997 [C(97)186/Final], and in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert.

Study elements conducted by subcontractors will be performed in accordance with the appropriate Standard Operating Procedures of the subcontractor.

E. Declaration of Intent

This study may be submitted to the United States Food and Drug Administration (FDA), the EMEA, an Organization for Economic Cooperation and Development (OECD) member country, ~~and/or the Japanese Ministry of Health, Labor and Welfare (JMHLW).~~

REASON FOR CHANGE:

The reference to the Japanese regulations was removed to be consistent with the other studies being conducted at Calvert.

PROTOCOL SECTION CHANGE (Page 5 of 17):

II. Introduction**K. Sponsor Representative**

Gary Woodnutt, PhD
Senior Vice President, Research
Lpath Inc.
4025 Sorrento Valley Blvd
San Diego, CA 92121
Phone: (858) 678-0800 ext 112
Email: ~~rmatteo@lpath.com~~ gwoodnutt@lpath.com

REASON FOR CHANGE:

To correct the contact information for the Sponsor Representative.

PROTOCOL SECTION CHANGE (Page 6 of 17):

II. Introduction**L. Principal Investigator for Dosing Formulation Analyses**

~~Victor T. Johnson, Ph.D.~~ **Victor J. Johnson, Ph.D.**

PROTOCOL SECTION CHANGE (Page 8 of 17):

III. Methods and Materials

A. Test Article and Vehicle

5. *Shipment and Analysis*

Dose formulation samples will be shipped to (on dry ice) and analyses will be performed under the direction of:

~~Victor T. Johnson, Ph.D.~~ **Victor J. Johnson, Ph.D.**

REASON FOR CHANGE:

There was a typographical error in the middle initial of the Principal Investigator.

PROTOCOL SECTION CHANGE (Page 9 of 17):

III. Methods and Materials

A. Test Article and Vehicle

6. *Sample Batch Acceptance Criteria (1st paragraph)*

For a sample batch run to be acceptable, **the following assay acceptance criteria will be met;**

Calibration Curve Acceptance Criteria:

- **Accuracy of standards: 20% of nominal (25% near LLOQ)**
- **Precision of the standards: %CV value $\leq 20.0\%$**
- **At least 75% of the standards (minimum of 6 points) are required to meet these criteria. Failed standards will be removed from the analysis.**

QC Sample Acceptance Criteria:

- **Accuracy of QC samples: 20% of nominal (25% near LLOQ)**

- **Precision of the QCs: %CV value $\leq 20.0\%$**
- **At least 67% of all QC samples are required to meet these criteria. In addition, at least 50% of the QC samples at each level are required to meet these criteria.**

~~a minimum of 75% of the QC samples for LT3114 at each concentration must be within $\pm 10.0\%$ from the nominal value. In addition, a minimum of 75% of the back-calculated CS values for LT-3114 must fall within 10.0% of their nominal values.~~

REASON FOR CHANGE:

The sample batch criteria was changed to reflect the criteria used by the analytical lab.

SIGNATURES:



Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

Date 18 Aug 2014



Steven D. Abraham
Scientific Management
Calvert Laboratories, Inc.

Date 18 Aug 2014



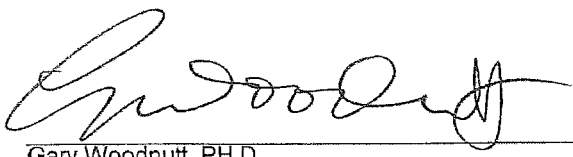
William M. Tuman
IACUC
Calvert Laboratories, Inc.

Date 18 Aug 2014



Marlene Modi, PH.D.
Study Monitor

Date 16 August 2014



Gary Woodnutt, PH.D.
Sponsor Representative

Date 18th August 2014



PROTOCOL AMENDMENT

STUDY NUMBER: 0200RL35.001
STUDY TITLE: Neuropharmacological Profile (NPP) of LT3114 in Rats (GLP)
AMENDMENT NUMBER: 2
EFFECTIVE DATE: 31 Oct 2014

PROTOCOL SECTION CHANGE (Page 9 of 17):

III. Methods and Materials

A. Test Article and Vehicle

5. Shipment and Analysis

Analytical Methods: To be added by protocol amendment **BRT 20140629-A**

Automated Data
Collection Systems: To be added by protocol amendment
Spectramax 340 microplate spectrophotometer and/or Spectramax M2e microplate spectrophotometer
Analysis of the ELISA absorbance data is performed with Softmax Pro v2.2.1 and/or v5.0.1

**Analysis of dose formulation
acceptance criteria is performed using
Microsoft Excel v2013**

**Dose formulation stability has been conducted under BRT
20140629-A**

REASON FOR CHANGE:

To add the Analytical Method and Automated Data Collection Systems and to include a reference to dose formulation stability.

PROTOCOL SECTION CHANGE (Page 9 of 17):

III. Methods and Materials

A. Test Article and Vehicle

6. Sample Batch Acceptance Criteria (2nd paragraph)

Concentration and homogeneity of dosing formulations will be determined by assay of formulation aliquots sampled from the top, middle and bottom locations of the dosing formulation container.

Measured concentration of the dosing formulation will be acceptable if it is ~~85%~~ **80%** to ~~115%~~ **120%** of the nominal concentration. Formulated LT3114 will be considered homogeneous if the measured concentrations of the top, middle and bottom aliquots have a %CV value ~~≤10.0%~~ **20%**.

REASON FOR CHANGE:

To update the dose formulation acceptance criteria

SIGNATURES:



Brian Klatt, B.S.
Study Director
Calvert Laboratories, Inc.

31 Oct 2014

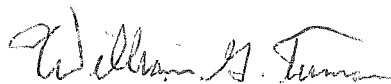
Date



Steven D. Shoben
Scientific Management
Calvert Laboratories, Inc.

31 Oct 2014

Date



William H. Turner
IACUC
Calvert Laboratories, Inc.

31 Oct 2014

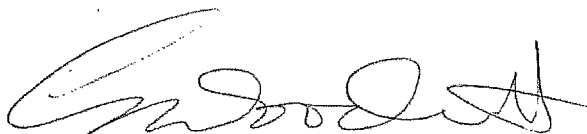
Date



Marlene Modi, PH.D.
Study Monitor

31 October 2014

Date



Gary Woodnutt, PH.D.
Sponsor Representative

31st October 2014

Date

Protocol Deviations

1. The relative humidity in the animal room deviated from the protocol-stated range of 30 to 70% on one occasion to a high of 76%. The room temperature in the animal room deviated from the protocol-stated range of 18 to 26°C on one occasion to a low of 19°C. The deviations were not of a magnitude or duration that would have affected the outcome of the study.
2. The strain/substrain listed in the protocol was CrI: CD®(SD) which is the stock designation for Charles River Laboratories. Since the animals were received from Harlan Laboratories, the stock designation was Hsd:SD. Since the correct strain/substrain of rats was used, this deviation had no impact on the outcome of the study.
3. The protocol stated that the dose formulation samples were to be shipped to the analytical laboratory on dry ice. The samples were stored refrigerated and were shipped on wet ice packs. Since the samples were sent under the correct storage condition, this deviation had no impact on the outcome of the study.